

KONA VENKATA SAI LAKSHMI

Senior Software Engineer | Java & Spring Boot | Cloud, Data Engineering & Distributed Systems

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Drive enterprise-scale digital transformation by architecting cloud-native applications, modernizing legacy platforms & delivering secure, scalable data, integration, and AI-enabled solutions across complex technology ecosystems.

EXECUTIVE SUMMARY

- Results-driven Senior Software Engineer with nearly 6 years of experience** in architecting & delivering cloud-native, event-driven & data-intensive platforms across telecom and enterprise ecosystems, leveraging Java, Spring Boot, Azure, Kafka, Databricks, and React.
- Proven expertise in enterprise-scale modernization initiatives**, transforming legacy CORBA, Struts, and monolithic applications into highly scalable microservices-based architectures, accelerating agility, maintainability, and operational resilience.
- Distinguished track record of building next-generation data platforms**, leading the design of Databricks-powered Enterprise Data Frameworks with CDC pipelines, Medallion Architecture, Delta Lake, and automated data governance capabilities.
- Adept at engineering secure, high-performance distributed systems, driving identity transformation** through Microsoft Entra ID, OAuth2/OIDC, JWT authentication, cloud security controls, and enterprise-grade observability frameworks.
- Passionate technology innovator with hands-on experience in AI-enabled software engineering**, integrating RAG-based solutions, LLM-powered automation, and intelligent developer productivity tools while championing DevOps, CI/CD, Agile, and SAFe delivery practices.
- Drove the design and development of scalable distributed systems and enterprise-grade UI applications**, enabling seamless integration, high availability, optimized performance, and robust user experiences across complex business ecosystems.

CORE COMPETENCIES

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| Enterprise Data Engineering & Architecture | Cloud-Native Microservices Architecture | Digital Transformation & Modernization |
| Distributed Systems & Platform Engineering | Data Platform Strategy & Engineering | Identity & Security Transformation |
| Technology Innovation & AI Adoption | Scalable Enterprise Application Development | Agile Delivery & Technical Leadership |

TECHNICAL SKILLS

- Languages and Backend:** Java 17/11/8, Python, SQL, Spring Boot, Spring Boot Actuator, JPA/Hibernate, Maven, REST APIs, Microservices
- Testing and Quality:** JUnit, Mockito, SonarQube, Code coverage gates, CI/CD quality checks, Code review
- Identity and Security:** OAuth2/OIDC, Microsoft Entra ID, JWT, JWKS validation, ECCUser session based authentication, Managed Identity, Sentry
- Data and Messaging:** Databricks, Delta Live Tables, Delta Lake, Medallion Architecture, CDC, SCD Type 1/2, GoldenGate, Apache Kafka, Confluent Cloud, Dead letter queues, IBM MQ, IE Bus
- Cloud and DevOps:** Microsoft Azure, AKS, Azure App Service, Docker, GitHub Actions, CI/CD, Redis/Redisson, Agile and Scrum, SAFe
- Frontend and AI:** React.js, JavaScript ES6+, HTML5, CSS3, Component based architecture, Responsive design, RAG agents, LLM integration
- Legacy Modernization:** CORBA, Apache Struts, ActiveVOS, BPEL, Sybase, DSAP, WebLogic, Tomcat, Oracle

WORK EXPERIENCE

Senior Associate Software Engineer | AT&T | Since Sep 2025

Key Roles and Responsibilities:

- Legacy Modernization & Event-Driven Transformation:** Organizing Dmaap legacy Struts-to-Spring Boot/Kafka modernization, transforming NRT services into event-driven microservices while integrating Entra authentication and Redis across Inquire & Manage services for enhanced scalability, security, performance, and real-time processing.
- Resilience, Alerting & Security Hardening:** Engineering asynchronous orchestration and alerting mechanisms while driving bug-bounty remediation, vulnerability hardening, and security-risk closure, strengthening resilience, compliance, and production readiness.
- Data Engineering Automation:** Engineering Python/YAML-driven BronzeTableFactory enabling zero-code source onboarding and establishing the enterprise-standard ingestion template.
- GenAI Engineering Intelligence:** Building RAG-based LLM agent for code-HLD validation, automated README/Mermaid documentation, and natural-language discovery of Kafka ownership and workflows.
- Digital Data Platform Transformation:** Driving enterprise-scale data platform transformation through Databricks, Delta Lake, and Azure, enabling scalable ingestion, processing, governance, and high-performance data ecosystems.
- Scalable Application & Integration Engineering:** Engineering cloud-native distributed applications using Java, Spring Boot, Kafka, and Confluent Cloud, while establishing resilient microservices and event-driven integration patterns.
- Secure Cloud & Intelligent Technology Enablement:** Strengthening cloud application security and intelligent capabilities through Azure, AKS, Entra ID, OAuth2/OIDC, JWT, and seamless Generative AI, RAG, and LLM integration.
- Enterprise Modernization & Technology Evolution:** Accelerating legacy-to-cloud modernization by decomposing monolithic platforms, implementing event-driven architectures, and embedding AI-powered solutions to drive scalability, automation, and operational agility.

Key Highlights:

- Architected Databricks CDC across 60+ tables and 8 schemas using 12 serverless DLT pipelines, Medallion Architecture, SCD Type 1/2, and automated data-quality quarantine.
- Modernized 20 DMAAP NRT modules to Kafka/Confluent Cloud with DLQs, replay, monitoring, Docker, CI/CD, caching, Managed Identity, and AKS.
- Migrated 15 applications to Entra ID & 10 Struts applications to Spring Boot with JWT/JWKS, JPA/Hibernate, automated testing, SonarQube & CI/CD gates.
- Enhanced Azure scalability and Redis/Redisson caching while remediating 30 high-priority vulnerabilities within 30 days.

Cloud Full Stack Developer | IBM | Jun 2021 – Aug 2025

- **Enterprise Integration Modernization:** Directed the migration of 30 DSM transaction types from CORBA to REST, engineering a bridge layer that routed CORBA stub invocations through REST APIs to Sybase, MQ, and DSAP backend.
- **Event-Driven Platform Transformation:** Led impact analysis across 80 DMAAP topics for DMAAP-to-Kafka modernization, identifying affected services and transforming NRT applications from DMAAP/Tomcat to Spring Boot with Kafka integration.
- **Workflow & Transaction Orchestration:** Engineered 15+ SCOPE and ActiveVOS workflow services using MQ-backed MDB listeners, triggering BPPEL processes, loading data into Oracle, and deploying BPR services on WebLogic; delivered billing and order-management capabilities spanning creation, generation, validation, and fulfillment.
- **Application Security & Performance Engineering:** Hardened legacy Struts applications against SQL injection, CSRF, and XSS, strengthened ECCUser session-based access controls, optimized database queries & connection pooling for high-throughput enterprise traffic & mentored junior engineers within Agile/Scrum delivery.

PROJECTS

AT&T | Sep 2025 – Oct 2026

- Accelerated DMAAP-to-Kafka migration by modernizing legacy Struts applications to Spring Boot microservices and implementing scalable Kafka message-consumption logic.
- Transformed ServiceNow-based contact retrieval into Spring Boot microservices with Redis caching, asynchronous processing, integrated dashboards, and proactive alerting; reduced downstream contact-update latency from 24 hours to <100ms across Kubernetes and Azure App Services.
- Remediated 10+ security vulnerabilities across enterprise applications, addressing critical risks including XSS, SQL Injection, CSRF, and other application-security threats.
- Architected Databricks and Microservices solutions and developed Databricks DLT pipelines implementing the Bronze-Silver-Gold Medallion Architecture, integrating data from Kafka, Oracle, and SFTP while leading a 12-member engineering team.
- Drove application modernization by migrating legacy Struts applications to Spring Boot services, improving scalability, maintainability, and architectural resilience.
- Led enterprise vulnerability remediation for applications hosted on Azure App Services, strengthening security posture and ensuring timely closure of identified vulnerabilities.
- Designed workflow orchestration using Temporal for the OFS project, enabling reliable, scalable, and resilient distributed process execution.
- Orchestrated delivery across 60+ cross-functional stakeholders spanning onsite and offshore/IDC teams, driving architecture alignment, execution governance, and seamless collaboration.

Personal Engineering Portfolio | React.js

- Developed & deployed production-grade React.js applications using functional components, Hooks and Context API for efficient state management.
- Implemented responsive CSS3 design and optimized npm-based build pipelines for performance and deployment efficiency.

IBM | Feb 2021 – Aug 2025

- Led DSM modernization by transforming monolithic web applications into CORBA and REST-based transactional services integrated with MQ, JMS, Sybase, and DSAP; implemented proactive alerts to identify downstream failures and service unavailability.
- Remediated 30+ security vulnerabilities across enterprise applications, addressing critical application-security risks including XSS, SQL Injection, CSRF, and related vulnerabilities.
- Drove DMAAP-to-Kafka migration by assessing modernization impacts across 20+ legacy applications and transforming message-consumption architecture from DMAAP to Kafka with Dead Letter Queue (DLQ) capabilities, strengthening resilience and fault handling.
- Delivered enterprise application enhancements for the SCOPE application, supporting workflow-driven ActiveVOS BPPEL/BPR services and optimizing business-process functionality.
- Demonstrated end-to-end Order-to-Cash workflows to stakeholders, showcasing the complete business lifecycle spanning Contracting, Ordering, and Billing modules and enabling clear functional alignment across teams.

AWARDS

- **Earned two Connection Awards & one Transformer Award at AT&T** within the first year of tenure, demonstrating exceptional performance & organizational impact.

EDUCATION & CERTIFICATION

- **2020:** B.Tech., Computer Science and Engineering, Vignan Institute of Information and Technology
- **Certifications:** Microsoft Certified: Azure Fundamentals (AZ-900) | SAE 5 Practitioner